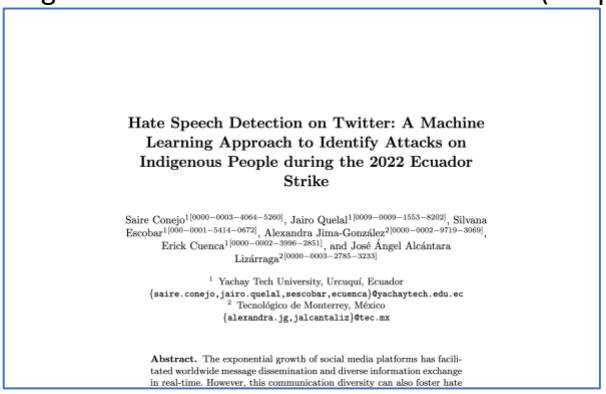
Anexo 1: Enlaces o evidencias a los artículos científicos

Nombre del grupo de investigación: Data Science and Analytics
Periodo del informe: 16-11-202 al 16-11-20223

- A Review of Techniques for Detecting Illicit Messages on Twitter. Román, S., Cuenca, E. ECTM 2023 - 2023 IEEE 7th Ecuador Technical Chapters Meeting, 2023 http://doi.org/10.1109/ETCM58927.2023.10309025

- Human Trafficking in Social Networks: A Review of Machine Learning Techniques. Bermeo, M., Escobar, S., Cuenca, E. Communications in Computer and Information Science, 2023, 1885 CCIS, pp. 22–36 http://doi.org/10.1007/978-3-031-45438-7_2

- Saire Conejo, Jairo Quelal, Silvana Escobar, Alexandra Jima, Erick Cuenca and Jose Alcántara. Hate Speech Detection on Twitter: A Machine Learning Approach to Identify Attacks on Indigenous People during the 2022 Ecuador Strike. AENIT2023 (Aceptado)



Hate Speech Detection on Twitter: A Machine Learning Approach to Identify Attacks on Indigenous People during the 2022 Ecuador Strike

Saire Conejo[1][0000−0003−4064−5260], Jairo Quelal[1][0009−0009−1553−8202], Silvana Escobar[1][000−0001−5414−0672], Alexandra Jima-González[2][0000−0002−9719−3069], Erick Cuenca[1][0000−0002−3996−2851], and José Ángel Alcántara Lizárraga[2][0000−0003−2785−3233]

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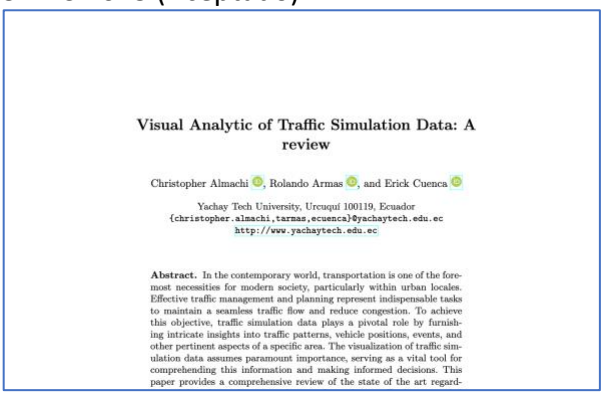
Abstract. The exponential growth of social media platforms has facilitated worldwide message dissemination and diverse information exchange in real-time. However, this communication diversity can also foster hate

- Christopher Almachi, Rolando Armas and Erick Cuenca Visual Analytic of Traffic Simulation Data: A review. ICSC-CITIES 2023 (Aceptado)



Visual Analytic of Traffic Simulation Data: A review

Christopher Almachi, Rolando Armas, and Erick Cuenca

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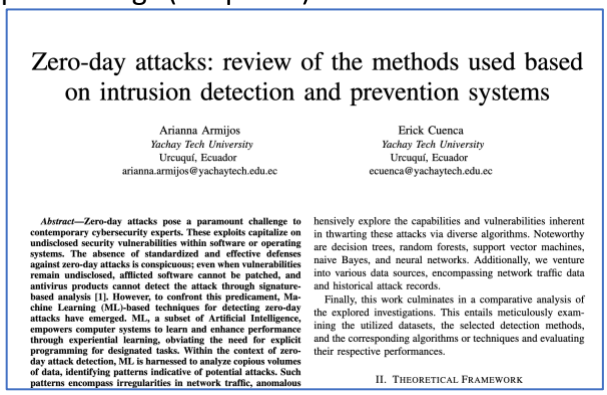
Abstract. In the contemporary world, transportation is one of the foremost necessities for modern society, particularly within urban locales. Effective traffic management and planning represent indispensable tasks to maintain a seamless traffic flow and reduce congestion. To achieve this objective, traffic simulation data plays a pivotal role by furnishing intricate insights into traffic patterns, vehicle positions, events, and other pertinent aspects of a specific area. The visualization of traffic simulation data assumes paramount importance, serving as a vital tool for comprehending this information and making informed decisions. This paper provides a comprehensive review of the state of the art regard-

- Arianna Armijos and Erick Cuenca. Zero-day attacks: review of the methods used based on intrusion detection and prevention systems download this paper signed view or update. IEEE C3 IEEE proceedings (Aceptado)



Zero-day attacks: review of the methods used based on intrusion detection and prevention systems

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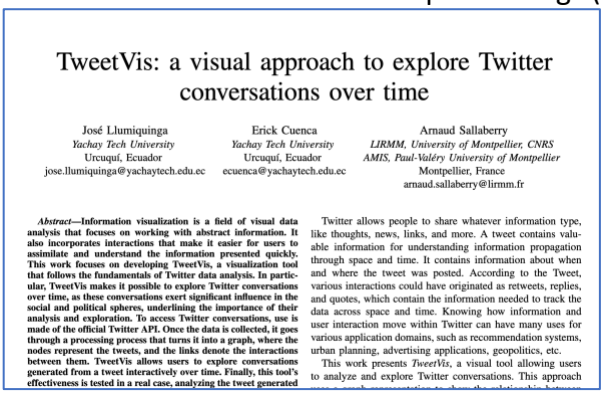
***Abstract*—Zero-day attacks pose a paramount challenge to contemporary cybersecurity experts. These exploits capitalize on undisclosed security vulnerabilities within software or operating systems. The absence of standardized and effective defenses against zero-day attacks is conspicuous; even when vulnerabilities remain undisclosed, afflicted software cannot be patched, and antivirus products cannot detect the attack through signature-based analysis [1]. However, to confront this predicament, Machine Learning (ML)-based techniques for detecting zero-day attacks have emerged. ML, a subset of Artificial Intelligence, empowers computer systems to learn and enhance performance through experiential learning, obviating the need for explicit programming for designated tasks. Within the context of zero-day attack detection, ML is harnessed to analyze copious volumes of data, identifying patterns indicative of potential attacks. Such patterns encompass irregularities in network traffic, anomalous**

hensively explore the capabilities and vulnerabilities inherent in thwarting these attacks via diverse algorithms. Noteworthy are decision trees, random forests, support vector machines, naive Bayes, and neural networks. Additionally, we venture into various data sources, encompassing network traffic data and historical attack records.

Finally, this work culminates in a comparative analysis of the explored investigations. This entails meticulously examining the utilized datasets, the selected detection methods, and the corresponding algorithms or techniques and evaluating their respective performances.

II. Theoretical Framework

- José Llumiquinga, Erick Cuenca and Arnaud Sallaberry. TweetVis: a visual approach to explore Twitter conversations over time. IEEE C3 IEEE proceedings (Aceptado)



TweetVis: a visual approach to explore Twitter conversations over time

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***Abstract*—Information visualization is a field of visual data analysis that focuses on working with abstract information. It also incorporates interactions that make it easier for users to assimilate and understand the information presented quickly. This work focuses on developing TweetVis, a visualization tool that follows the fundamentals of Twitter data analysis. In particular, TweetVis makes it possible to explore Twitter conversations over time, as these conversations exert significant influence in the social and political spheres, underlining the importance of their analysis and exploration. To access Twitter conversations, use is made of the official Twitter API. Once the data is collected, it goes through a processing process that turns it into a graph, where the nodes represent the tweets, and the links denote the interactions between them. TweetVis allows users to explore conversations generated from a tweet interactively over time. Finally, this tool's effectiveness is tested in a real case, analyzing the tweet generated**

Twitter allows people to share whatever information type, like thoughts, news, links, and more. A tweet contains valuable information for understanding information propagation through space and time. It contains information about when and where the tweet was posted. According to the Tweet, various interactions could have originated as retweets, replies, and quotes, which contain the information needed to track the data across space and time. Knowing how information and user interaction move within Twitter can have many uses for various application domains, such as recommendation systems, urban planning, advertising applications, geopolitics, etc.

This work presents *TweetVis*, a visual tool allowing users to analyze and explore Twitter conversations. This approach

- Harvey Marin, Erick Cuenca, Diego Morales-Navarrete and Ruben Basantes. Machine Learning Applied to the Analysis of Glacier Masses. TICEC 2023. http://doi.org/10.1007/978-3-031-45438-7_11

- Mike Bermeo, Manuel Eugenio Morocho-Cayamcela and Erick Cuenca Unraveling the power of 4D residual blocks and transfer learning in violence detection. TICEC 2023 http://doi.org/10.1007/978-3-031-45438-7_14

- Stance-EC: A Spanish-language Dataset of Polarized Twitter Conversations for Stance Detection. Leo Ramos, Mike Bermeo, Silvana Escobar, Diego Morales-Navarrete, and Erick Cuenca (Enviado a revisión)

Stance-EC: A Spanish-language Dataset of Polarized Twitter Conversations for Stance Detection

Stance-EC: A Spanish-language Dataset of Political Discourse on Twitter for Stance Detection

Leo Ramos, *Member, IEEE,* Mike Bermeo, *Student Member, IEEE,*
Silvana Escobar, Diego Morales-Navarrete, and Erick Cuenca, *Senior Member, IEEE*

***Abstract*—Stance detection is a task in natural language processing that involves identifying and categorizing opinions in text. It has various applications, such as improving our understanding of socio-political discussions and informing market analysis. Social media, especially Twitter, is crucial for social discourse. However, using Twitter for stance detection faces a significant challenge: the lack of reliable language-specific datasets. While English datasets for stance detection are prevalent, Spanish-language resources remain limited. In this paper, we present Stance-EC, a new Spanish-language dataset for stance detection. We aim to provide a valuable resource with polarized content,**

I. Introduction

Social media has become an integral part of the modern world, allowing people to share their thoughts and opinions on various topics with a global audience [1], [2]. One of the most popular social media platforms is Twitter [3], [4], which has been widely used for discussions and debates on various topics, including politics, social issues, and current events [5], [6].

Its prominence as a source for real-time sentiments, con-